

## STATES OF MATTER

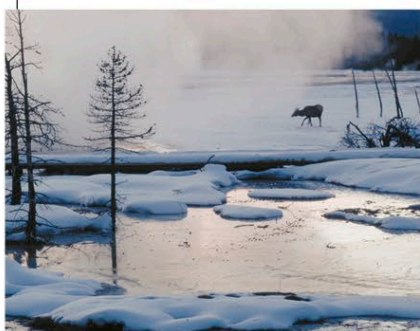
Ordinary matter exists in four states, called solid, liquid, gas, and plasma. These differ in the energy of the matter's particles (molecules, atoms, or ions) and in the particles' freedom to move relative to one another. Substances can transfer between states—by losing or gaining heat energy, for instance. The constituents of a solid are locked by strong bonds and can hardly move, whereas in a liquid they are bound only by weak bonds and can move freely. In a gas, the particles are bound very weakly and move with greater freedom, occasionally colliding.

A gas becomes a plasma when it is so hot that collisions start to knock

electrons out of its atoms. A plasma therefore consists of ions and electrons moving extremely energetically. Because stars are made of plasma, it is the most common state of ordinary matter in the universe; the gaseous state is the second most common.

### SOLID, LIQUID, AND GAS

On Earth, water can sometimes be found as a liquid, in solid form (ice or snow), and as a gas (water vapor), all in close proximity.



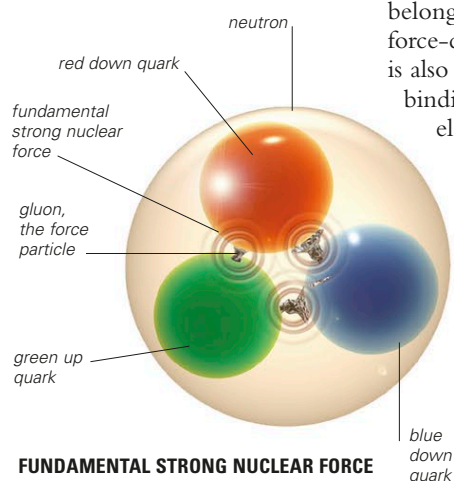
## FORCES INSIDE MATTER

The bonds that hold solids, liquids, gases, and plasma together are based on the electromagnetic (EM) force. This force attracts particles of unlike electrical charge. It is one of three forces that control the small-scale structure of matter. The others are the strong nuclear force, composed of fundamental and residual parts; and the weak nuclear force or interaction. Together with a fourth force, gravity, these are the fundamental forces of nature. The EM, weak, and strong forces are mediated by force-carrier particles, which

belong to a group of particles called the bosons. (A force-carrier particle for gravity, called the graviton, is also hypothesized.) The EM force, as well as binding atoms in solids and liquids, also holds together protons, neutrons, and atomic nuclei. The weak force brings about radioactive decay and other nuclear interactions.

### PLASMA

Plasma exists naturally in stars but can also be artificially created. In a plasma ball, electricity is induced to flow from a charged metal ball through a gas to the surface of a glass sphere, creating plasma streamers.

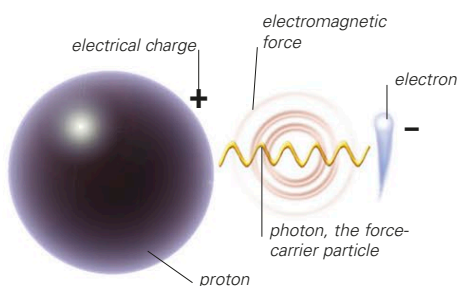
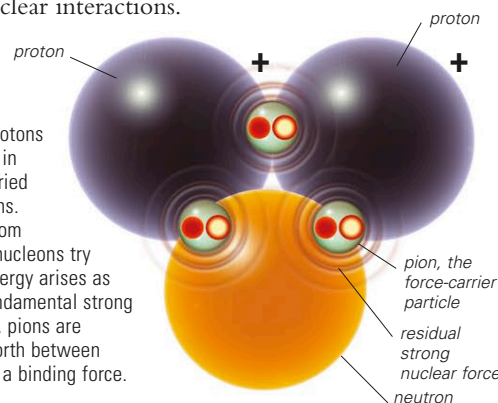


### FUNDAMENTAL STRONG NUCLEAR FORCE

Also known as the color force, this force binds quarks within protons and neutrons. It controls the quarks' "color" property, and as it operates, the quarks constantly change "color" by exchanging virtual gluons (the force-carrier particles).

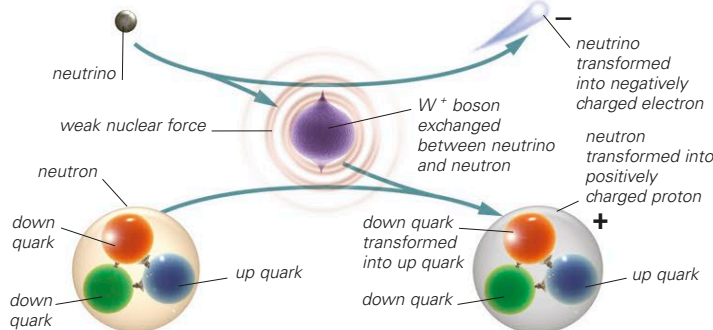
### RESIDUAL STRONG NUCLEAR FORCE

This force binds the protons and neutrons together in atomic nuclei. It is carried by particles called pions. Pions are generated from energy created when nucleons try to move apart. This energy arises as a by-product of the fundamental strong force. Once generated, pions are exchanged back and forth between the nucleons, creating a binding force.



### ELECTROMAGNETIC FORCE

Within an atom, the electromagnetic (EM) force holds the electrons within the shells surrounding the nucleus. It attracts the negatively electrically charged electrons toward the positively charged nucleus and keeps electrons apart. The force carrier for the EM force is the photon.



### WEAK INTERACTION, OR WEAK NUCLEAR FORCE

This force governs radioactive decay, among other interactions. Its force-carrier particles are  $W^+$ ,  $W^-$ , and  $Z^0$  bosons. The interaction illustrated above is called neutrino-neutron scattering. Here, a  $W^+$  boson controls the changing of a neutrino into an electron and the transformation of a down quark into an up quark, converting a neutron into a proton.

## STEVEN WEINBERG

American physicist Steven Weinberg (b. 1933) is best known for his theory that two of the fundamental forces—the weak interaction and the electromagnetic force—are unified, or work in an identical way, at extremely high energy levels, such as those existing just after the Big Bang (see p.48). Weinberg's so-called electroweak theory was confirmed by particle accelerator experiments in 1973. He and his colleagues received the 1979 Nobel Prize for physics.

